

AMTI (NMTC) - 2012

RAMANUJAN CONTEST - FINAL - INTER LEVEL

1. Find all the pairs (x, y) where x, y are integers satisfying $(2x - 1)^3 + 16 = y^4$.
2. I is the incentre of the isosceles triangle ABC in which $AB = AC$. Let Σ be a circle which touches AB at E and AC at F and touches the circumcircle of triangle ABC internally. Prove that I lies on EF .
3. A function $f : \mathbb{Q} \rightarrow \mathbb{Q}$, where $\mathbb{Q}$ is the set of rational numbers, satisfies the conditions
 - (a) $f(1) = 2$.
 - (b) $f(xy) + f(x + y) = f(x) \cdot f(y) + 1$ for all $x, y \in \mathbb{Q}$. Determine all such functions f , with proof.
4. Let B be a point on the circle Σ_1 and A be a point on the tangent at B to Σ_1 ($B \neq A$). Let C be a point not on Σ_1 such that AC meets Σ_1 in two distinct points. Let Σ_2 be a circle touching AC at C and Σ_1 and D on the same side of AC as B . Prove that the circumcentre of triangle BCD lies on the circumcircle of triangle ABC .
5. Find all positive integers x, y, z such that $8^x + 15^y = 17^z$.
6. A Pythagorean triangle is a right triangle in which all the three sides are of integer lengths. Let a, b be the legs of a Pythagorean triangle, and h be altitude to the hypotenuse. Determine all such triangles for which $\frac{1}{a} + \frac{1}{b} + \frac{1}{h} = 1$.
7. Let $f(n)$ be a function defined on the non-negative integer n . Given
 - (a) $f(0) = f(1) = 0$
 - (b) $f(2) = 1$
 - (c) for $n > 2$, $f(n)$ gives the smallest positive integer which does not divide n .
8. In a circle C with centre O and radius r , let C_1 and C_2 be two circles with centres O_1, O_2 and radii r_1 and r_2 respectively be situated such that each circle C_1 and C_2 is internally tangent to C at A_1 and A_2 respectively and such that C_1 and C_2 are externally tangent to each other at A . Prove that the three lines OA, O_1A_2 and O_2A_1 are concurrent.
